

Clearinghouse API & Healthcare Connectivity

Cost · Coverage · Limitations · Product Opportunity

Can a clearinghouse API give us most of the data we need — what does it cost, what does it *not* cover, and where do we still need direct payer APIs, FHIR, portals, AI voice and humans?

PREPARED FOR	AI Revenue Recovery OS
DATE	August 20, 2026
SCOPE	Stedi · Availity · Waystar · Optum/Change · Experian · Office Ally · Claim.MD · TriZetto
MVP CONNECTIVITY (est.)	~\$312/mo on free/sandbox tiers
CLASSIFICATION	Confidential · Investor & technical · Sourced 2026 · MODEL ASSUMPTIONS where noted

Disclosures & method

Pricing is researched 2026 with sources; undisclosed pricing is recorded **“QUOTE REQUIRED — not publicly disclosed”** and never invented. Free sandbox is never equated with free production; direct payer API is never equated with clearinghouse connectivity; payer connectivity is never equated with full payer functionality. Connectivity cost figures are **MODEL ASSUMPTIONS** (per-transaction clearinghouse rates are usage-tiered / quote-only). HIPAA-aligned architecture — not “HIPAA certified.” Final medical-necessity/coverage decisions require a qualified clinician (CMS-4201-F; CA SB 1120). No PHI.

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1. Executive Summary

A modern healthcare clearinghouse API can deliver **most of the standardized transaction data** the platform needs — eligibility, benefits, claim submission, claim status and electronic remittance — through a single integration reaching thousands of payers. It **cannot**, on its own, deliver prior-authorization for every payer, appeals (no universal standard), payer medical policies, or the reasoning that turns a denial into a recovered dollar. Those are exactly the layers we build.

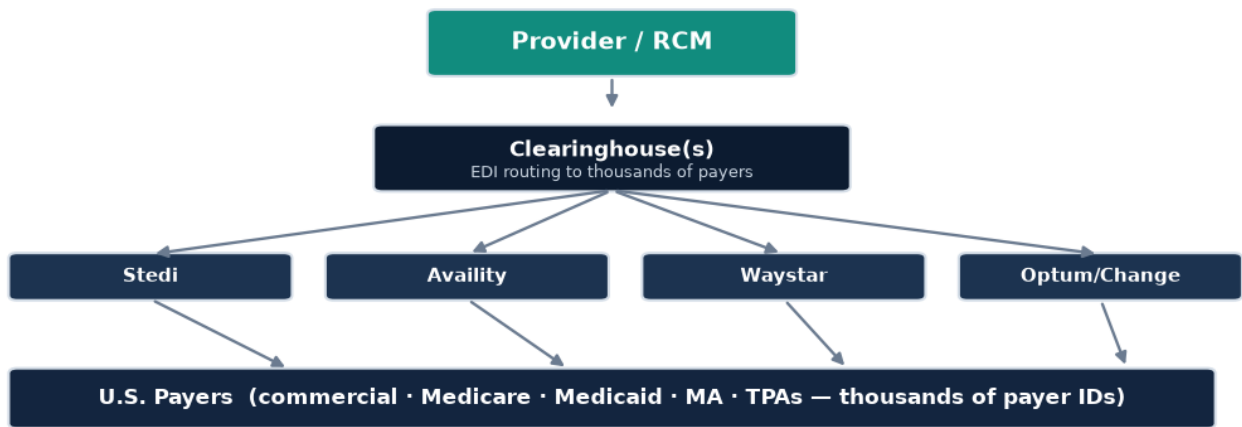
Question	Answer
Is a clearinghouse API free?	Partly. Free SANDBOX and small free tiers exist (Stedi Basic 100 txns/mo; Availity Essentials portal). Free PRODUCTION at scale generally does not — most pricing is usage-based or QUOTE REQUIRED.
What can clearinghouses cover?	Eligibility (270/271), claims (837), claim status (276/277), ERA (835), acknowledgments (277CA/999/TA1); attachments (275) emerging.
What can't they cover?	Universal prior auth, appeals, payer medical policies, denial reasoning, patient-estimate logic, cross-channel orchestration.
How many payers?	Thousands via one integration (Stedi 3,500+; Availity 4,000+; TriZetto 8,000+ connections) — but connectivity ≠ full functionality.
Do we still need payer APIs / FHIR / portals / voice?	Yes — selectively. Direct payer APIs & FHIR (2027) for richer/PA data; portals, AI voice and humans as fallback.
MVP connectivity cost	~\$312/month on free/sandbox tiers (near-\$0).
Production connectivity (500k claims/mo)	~\$284,000/month (MODEL ASSUMPTION; verify live rates).

Bottom line

Do not build a clearinghouse. Build an **AI Revenue Recovery orchestration platform** that uses clearinghouses as ONE connectivity layer — alongside direct payer APIs, FHIR, portals, AI voice and human workflows — governed by an AI connectivity orchestrator. Start provider-side on clearinghouse APIs (near-\$0 MVP), and invest the proprietary effort in payer intelligence, denial/appeal reasoning and orchestration.

2. What Is a Healthcare Clearinghouse?

A clearinghouse is an intermediary that connects providers to payers. It accepts a provider's transaction (e.g. an eligibility inquiry or a claim), validates and formats it to each payer's requirements, routes it to the correct payer, and returns the response. It abstracts away thousands of payer-specific connections, formats and enrollment processes behind one integration.



Payer connectivity ≠ full payer functionality — capability varies by payer & contract

Direct payer APIs · FHIR (2027) · Portals · Voice · Human sit **ALONGSIDE** the clearinghouse, not replaced by it

Figure 1. The clearinghouse sits between the provider and thousands of payers — but direct payer APIs, FHIR, portals, voice and humans sit alongside it, not replaced by it.

Sources: X12 Health Care Transaction Flow (High) · CMS Adopted Standards & Operating Rules (High)

3. Clearinghouse vs Payer vs API vs FHIR vs Portal vs Voice vs Human

These are not interchangeable. Conflating them is the most common and most expensive mistake in RCM connectivity planning.

Term	What it is	When it's the right channel
Clearinghouse	Intermediary routing EDI to many payers	Standard transactions at scale (eligibility, claims, status, ERA)
Payer	The insurer that adjudicates	N/A — the endpoint, not a channel
Payer API	A direct programmatic interface to one payer	When a payer offers richer/real-time data than the clearinghouse relays
FHIR	HL7 REST standard (SMART/Da Vinci)	Prior auth (PAS/CRD/DTR) & CMS-regulated payer data (2027)
EDI / X12	The mandated transaction format (270/837/835...)	The substrate under most clearinghouse & payer transactions
Portal	A payer's web UI for humans	When no EDI/API/FHIR path exists (auth follow-up, attachments, appeals)
Voice	Phone call to payer IVR/rep (AI or human)	Only when no digital channel resolves it
Human	A person doing the work	Judgment, exceptions, complex appeals — last resort

Three distinctions we never blur

① Free **sandbox** ≠ free **production**. ② Direct **payer API** ≠ **clearinghouse** connectivity. ③ Payer **connectivity** ≠ full payer **functionality**.

4. Is a Clearinghouse API Free?

Short answer: **parts are; production at scale generally is not.** There are genuinely free developer sandboxes and small free production tiers, but most vendors price production transactions on usage or by quote.

- **Genuinely free (with limits):** Stedi Basic plan — 100 free production transactions/month (eligibility, claims, ERA, claim status), processed manually in the portal. Availity Essentials portal — free to providers for sponsored/participating payers.
- **Free sandbox / developer:** Stedi test-mode keys (mock data, fixed payers); Availity Demo plan (3 TPS / 500 per day). A free developer account is NOT free production.
- **Usage-based / quote:** Stedi pay-as-you-go (tiered, exact rates QUOTE REQUIRED); Availity/Optum/Waystar/Experian/TriZetto API pricing QUOTE REQUIRED.
- **Transparent low-cost:** Office Ally — free to participating payers, \$44.95/month per Tax ID+NPI for non-participating; Claim.MD — published monthly tiers (~\$30/\$60/\$120).

Sources: Stedi Basic plan (100 free txns/mo) (High) · Stedi pay-as-you-go pricing (High) · Availity Essentials (free portal) (High) · Office Ally Par vs Non-Par fees (\$44.95) (High)

5. Free Sandbox vs Paid Production

The distinction that most often misleads planning: a free developer account or sandbox says nothing about production cost. This matrix keeps them separate.

Provider	Sandbox	Dev acct	Production	Setup	Monthly	Txn fee	Notes
Stedi	YES (test keys)	YES (self-serve)	YES	None	None (PAYG)	Tiered PAYG — QUOTE	Basic plan: 100 free production txns/mo; prepaid from \$100
Availity	YES (Demo plan)	YES	YES	QUOTE	Essentials free (par payers)	QUOTE	Free provider portal for sponsored payers; API pricing quote-only
Waystar	Partner-gated	Partner only	YES	QUOTE	QUOTE	QUOTE	Enterprise/contract; no public sandbox
Optum/Change	YES (rep-issued)	Gated	YES	QUOTE	QUOTE	QUOTE	Dev portal but contract-gated
Experian Health	UNKNOWN	No public	YES	QUOTE	QUOTE	QUOTE	Enterprise; 900+ payers
Office Ally	UNKNOWN	No public API	YES	None (Par)	None (Par)	\$0 Par / \$44.95 Non-Par/mo	Par payers free; Non-Par monthly fee
Claim.MD	UNKNOWN	YES (docs)	YES	None	~\$30–\$120/mo	Tier-inclusive	Transparent monthly tiers (confirm official)
TriZetto	Enterprise	No public	YES	QUOTE	QUOTE	QUOTE	Enterprise; 8,000+ connections

6. Major Clearinghouses

Eight clearinghouses were researched. They split into **developer-native** (Stedi, Claim.MD), **network + developer program** (Availity, Optum/Change), and **enterprise/contract** (Waystar, Experian, TriZetto), plus a transparent-priced web/SFTP option (Office Ally).

Clearinghouse	Type	Self-serve API	Pricing	Payers
Stedi	API-first clearinghouse (developer-native)	YES	Public	3,500+ payers (aggregate)
Availity	Network/clearinghouse + free provider portal + developer program	YES	QUOTE	4,000+ payers
Waystar	Enterprise RCM platform (clearinghouse + AI)	Gated	QUOTE	5,000+ payer connections (varies by page)
Optum / Change Healthcare	Largest clearinghouse (~50% of U.S. medical claims) + developer marketplace	Gated	QUOTE	~2,400 payers (Optum Medical Network/EDI page)
Experian Health	Enterprise RCM + data (eligibility, discovery, AI denials)	NO	QUOTE	900+ U.S. payers/MCOs/TP As/government (direct connections)
Office Ally	Web/SFTP clearinghouse + free practice tools	NO	Public	'Hundreds' of payers (exact count UNKNOWN)
Claim.MD	Low-cost transparent clearinghouse (small/solo/billing cos)	YES	Public	~2,800+ payers (secondary)
TriZetto Provider Solutions (Cognizant)	Enterprise clearinghouse (650+ PM/EHR interfaces)	NO	QUOTE	8,000+ payer connections

Sources: Stedi pricing / docs / billing (High) · Availity Developer Portal getting started (High) · Waystar platform (financial clearance/denials) (High) · Optum developer — access the APIs (High) · Experian Health eligibility (900+ payers) (High) · Office Ally pricing (Par/Non-Par) (High) · Claim.MD quick-start / pricing (Med) · TriZetto Provider Solutions clearinghouse (Med)

7. Stedi — Deep Dive

Dimension	Detail
Type	API-first clearinghouse (developer-native)
Self-serve / access	YES — public docs, self-serve sign-up, MCP/Agent tooling
Sandbox	YES — free test-mode keys; mock 270/271 for a fixed payer set (Aetna, Cigna, UHC, CMS); cannot submit test claims via API/SFTP (production account required)
What's free	Basic plan: 100 free PRODUCTION transactions/month (eligibility, claims, ERA, claim status each), processed manually in the portal
Pricing	Pay-as-you-go, 100% usage-based tiered (per-unit price falls with volume); no monthly minimum, no setup fee, no per-provider/per-payer fees; prepaid balance from \$100. Exact per-transaction rates: QUOTE REQUIRED — not publicly disclosed (interactive estimator only).
Transactions (API)	270/271, 276/277 (real-time), 837P/I/D, 835, 275 attachments, 277CA, 999
Prior authorization	No dedicated 278 API; surfaces PA REQUIREMENT via 271 authOrCertIndicator only
Payer network	3,500+ payers (aggregate); one-click enrollment for 850+
Enrollment	ERA always requires enrollment; claims/eligibility payer-dependent; no Stedi enrollment fees
Security	HIPAA-covered clearinghouse; signed BAA required for production PHI; SOC 2 Type II; HIPAA-eligible (audited)
Confidence	High

Sources: Stedi pricing / docs / billing (High)

8. Availity — Deep Dive

Dimension	Detail
Type	Network/clearinghouse + free provider portal + developer program
Self-serve / access	YES — Developer Portal: Demo plan auto-approved; Standard (production) via subscription + Contact Sales
Sandbox	YES — Demo plan sandbox (3 TPS / 500 txns per day; OAuth2 required)
What's free	Availity Essentials portal free to providers for SPONSORED/participating payers (no setup/monthly fee); non-sponsoring payers need paid Essentials Plus
Pricing	API/production pricing QUOTE REQUIRED — not publicly disclosed. Rate tiers published (Demo 3 TPS/500 per day; Standard 100 TPS/100,000 per day) but not dollar prices.
Transactions (API)	270/271 (Coverages API), 276/277 (Enhanced Claim Status), 837 (claims), 835 (ERA), 278 (Service Reviews), Claim Attachments API; FHIR prior-auth; CORE-certified
Prior authorization	YES — Service Reviews API (278) + IsAuthRequired + Auth Attachments; end-to-end FHIR prior auth; 'Intelligentum' AI
Payer network	4,000+ payers; 95%+ of payers; 3.5M+ providers; 17,000+ trading partners; ~8.8M daily txns
Enrollment	Payer registration/agreements via Trading Partner Management
Security	HIPAA covered entity/BA; OAuth2 over HTTPS; CORE-certified (2024)
Confidence	High

Sources: Availity Developer Portal getting started (High)

9. Waystar — Deep Dive

Dimension	Detail
Type	Enterprise RCM platform (clearinghouse + AI)
Self-serve / access	NO — REST/batch APIs exist but partner/contract-gated; no public self-serve developer portal or sandbox
Sandbox	Partner-gated (no public sandbox)
What's free	None (enterprise/contract)
Pricing	QUOTE REQUIRED — not publicly disclosed (custom/contract). Third-party estimates exist but are NOT official.
Transactions (API)	Eligibility, 837 claims (API or SFTP batch), claim status, 835 ERA, denial/appeal mgmt, prior auth, coverage discovery
Prior authorization	YES — Authorization Suite (Digitize.AI); rules engine checks ~6.8M payer rules
Payer network	5,000+ payer connections (varies by page); 1M+ providers; 100,000+ live integrations; \$200B+ payments/yr
Enrollment	Enrollment + payer agreements + EHR/PM integration via implementation services
Security	Enterprise HIPAA/BA; contract-based
Confidence	High(quote)/Med(figures)

Sources: Waystar platform (financial clearance/denials) (High)

10. Optum / Change Healthcare — Deep Dive

Dimension	Detail
Type	Largest clearinghouse (~50% of U.S. medical claims) + developer marketplace
Self-serve / access	Dev portal + sandbox (developer.optum.com / marketplace.optum.com) but access issued by Optum rep / contract
Sandbox	YES — sandbox access via Optum rep
What's free	None self-serve (contract)
Pricing	QUOTE REQUIRED — not publicly disclosed
Transactions (API)	270/271 (Enhanced Eligibility), 276/277 (Claim Status v2), 837P/I, 835, 278 (Prior Auth Submission API, real-time), 275 (Attachments API), Coverage Discovery, MBI lookup, COB
Prior authorization	YES — Prior Authorization Submission API (real-time)
Payer network	~2,400 payers (Optum Medical Network/EDI page)
Enrollment	Payer-specific transaction enrollment (837/835/EFT); fees not public
Security	HIPAA; post-2024-breach services operational under Optum brand
Confidence	High(caps)/Med(figures)

Sources: Optum developer — access the APIs (High)

11. Other Clearinghouses

Experian Health

Dimension	Detail
Type	Enterprise RCM + data (eligibility, discovery, AI denials)
Self-serve	NO public self-serve program (enterprise/contract; EDI + EHR connectors)
Pricing	QUOTE REQUIRED — not publicly disclosed (volume/complexity-based)
Transactions	270/271 (real-time), 276/277 (Enhanced Claim Status), 837 (ClaimSource), 835, MBI lookup, COB (CAQH), insurance/coverage discovery (Wave HDC)
Payers	900+ U.S. payers/MCOs/TPAs/government (direct connections)

Office Ally

Dimension	Detail
Type	Web/SFTP clearinghouse + free practice tools
Self-serve	No prominent public developer API (web + SFTP; enterprise EDI gateway custom)
Pricing	Non-Par payers: \$44.95/month per unique Tax ID + Rendering NPI in any month Non-Par claims are sent (criteria updated eff. 2026-01-01). EHR ~\$39.95–\$44.95/provider/mo (secondary).
Transactions	837P/I/D, real-time 270/271, 835 ERA, claim status
Payers	'Hundreds' of payers (exact count UNKNOWN)

Claim.MD

Dimension	Detail
Type	Low-cost transparent clearinghouse (small/solo/billing cos)
Self-serve	YES — public developer docs (docs.claim.md); API quick-start for claims/eligibility/ERA
Pricing	Published monthly tiers (secondary; confirm on official page): Basic ~\$30/mo; Small Volume ~\$60/mo (100 claims + 100 ERA + 100 eligibility); Unlimited ~\$120/mo (unlimited claims + ERA + 1,000 eligibility). Per-claim option referenced (~\$0.10–\$0.30) but inconsistent → treat as UNKNOWN.
Transactions	837 claims, 270/271 eligibility, 835 ERA, attachments, claim status
Payers	~2,800+ payers (secondary)

TriZetto Provider Solutions (Cognizant)

Dimension	Detail
Type	Enterprise clearinghouse (650+ PM/EHR interfaces)
Self-serve	Enterprise/contract-only; 'Unify' headless API (AI agents as consumers) emerging
Pricing	QUOTE REQUIRED — not publicly disclosed. Secondary ~\$0.15–\$0.40/claim estimate is UNVERIFIED.
Transactions	837P/I/D, 835 (incl. EOB→835 conversion), 270/271, 276/277, prior auth/denials workflow
Payers	8,000+ payer connections; 875,000+ providers; ~4.4B annual transactions

12. EDI Transaction Matrix

The mandated X12 transactions and how reachable each is. Acknowledgments (277CA/999/TA1) are transport-level; 278 (prior auth) and 275 (attachments) are the least standardized.

X12	Purpose	CH support	API	Payer dependency	Free sandbox
270	Ask a payer if a patient is covered and what the benefits are	YES — universal	YES (most CH APIs)	Low — HIPAA-mandated standard	YES (most)
271	The payer's answer to a 270 — active/inactive, plan, benefits	YES — universal	YES	Low	YES
276	Ask a payer what happened to a submitted claim	YES — common	YES (most CH APIs)	Medium — payer variability	YES (most)
277	The payer's answer to a 276 — pending/paid/denied/in review	YES — common	YES	Medium	YES
277CA	Confirms a claim was accepted or rejected at intake (front-end edits)	YES — common	PARTIAL (via CH)	Low-Medium	YES
278	Request approval before a service and get the payer's decision	PARTIAL — payer-dependent	PARTIAL	HIGH — least standardized	PARTIAL
837P	Submit a physician / professional services claim	YES — universal	YES (most CH APIs)	Low — HIPAA-mandated	YES (test)
837I	Submit a hospital / facility claim	YES — universal	YES	Low	YES (test)
837D	Submit a dental claim	YES — common	PARTIAL	Low	YES (test)
835	The payer's electronic explanation of payment and adjustments	YES — universal	YES (most CH APIs)	Low	YES (test)
999	Confirms an EDI file was syntactically accepted or rejected	YES — universal	N/A (transport)	None — transport layer	YES
TA1	Confirms the EDI interchange envelope was received/valid	YES — universal	N/A (transport)	None — transport	YES
275	Send supporting documentation (records, notes) to a payer	PARTIAL — emerging	PARTIAL (CH/FHIR)	HIGH — payer-dependent	PARTIAL

Sources: X12 Health Care Transaction Flow (High) · CMS Adopted Standards & Operating Rules (High)

13. Eligibility (270/271)

Eligibility is the most-adopted transaction (96% electronic). A 271 reliably confirms active coverage and plan, but detailed benefit accumulators vary widely by payer. The field matrix below shows what is typically returned.

Eligibility (270 → 271)



271 field	Availability	Note
Active / inactive coverage	YES	Core EB segment; near-universal
Effective date	YES	Usually present
Termination date	PARTIAL	Present when coverage ended/known; often blank if active
Member ID	YES	Echoed/normalized in 271
Plan / group	YES	Plan name/number typically returned
Product (HMO/PPO/EPO)	PARTIAL	Sometimes in plan description; not a discrete field for all payers
Network status	PAYER DEPENDENT	In/out-of-network benefit differentiation varies
Deductible (individual/family)	PARTIAL	Common for many payers; not guaranteed
Deductible remaining	PAYER DEPENDENT	High-value but inconsistently returned
Copay	PARTIAL	Service-type dependent; often returned for common service types
Coinsurance	PARTIAL	Service-type dependent
Out-of-pocket maximum	PAYER DEPENDENT	Returned by some payers
Out-of-pocket remaining	PAYER DEPENDENT	Inconsistent; high-value when present
Service-specific benefits	PAYER DEPENDENT	Depends on service type codes requested (EQ) and payer support
Authorization/referral requirement	PAYER DEPENDENT	Some 271s flag 'auth required'; not reliable across payers

Sources: 2025 CAQH Index (adoption / savings) (Med-High) · Stedi pricing / docs / billing (High)

14. Benefits

Benefit detail is returned in the 271 but completeness is **payer-dependent**. Do not assume every payer returns deductible-remaining, OOP-remaining or service-specific copay/coinsurance. Richer benefit data sometimes requires the payer's own API/portal or FHIR.

Channel	Benefit data realistically obtainable	Limitation
270/271 (EDI/API)	Active coverage, plan, common copay/coinsurance, some accumulators	Field completeness varies by payer; EQ service-type dependent
Payer API	Sometimes richer real-time benefits	Only where a payer offers it; not universal
FHIR	Coverage/benefit resources (emerging; 2027 for CMS payers)	Early adoption; patient-authorized for Patient Access
Portal	Full benefit detail as shown to staff	Manual / RPA; brittle

15. Patient Responsibility

A clearinghouse generally provides the **data to calculate** an estimate — not a final patient-responsibility figure. Our platform computes the estimate from allowed amount, deductible remaining, copay, coinsurance and OOP remaining, combined with the provider's contracted/fee-schedule amounts.

Key distinction

Clearinghouse = insurance/benefit DATA. Our platform = the ESTIMATE. Final patient responsibility is only known after adjudication (835). Estimation quality depends on benefit-field completeness (payer-dependent) and contracted rates (ours).

16. Claims (837P / 837I / 837D)

Claim submission is a core, universal clearinghouse function via API, SFTP, EDI, portal, batch or real-time. Payer enrollment may be required for specific payers. Pricing is per-claim, bundled or tiered — and QUOTE REQUIRED for most enterprise vendors.

Claim submission (837 → 277CA → 835)



Channel	Supported	Note
API	YES (Stedi, Availity, Optum, Claim.MD)	Real-time or async
SFTP / batch	YES (most)	High-volume nightly batches
EDI direct	YES	Underlying format
Portal	YES	Manual entry fallback

17. Claim Status (276/277)

Claim status (81% electronic) is widely available real-time or batch via clearinghouse APIs. Status granularity varies — some payers return rich pending/denied/paid detail with medical-review flags; others return minimal status. This is a prime AI-voice fallback when the 277 is unhelpful.

Claim status (276 → 277)



Sources: 2025 CAQH Index (adoption / savings) (Med-High) · Availity HIPAA transactions (APIs) (High) · Optum developer — access the APIs (High)

18. ERA / Payment (835)

The 835 is the payer's electronic explanation of payment: what was paid, adjustments, CARC/RARC codes, patient responsibility and the trace/EFT reassociation number. ERA enrollment is usually required per payer. The 835 is the primary fuel for our denial-intelligence engine.

835 element	Feeds
Payment & adjustment amounts	Underpayment detection
CARC / RARC codes	Denial reason & root-cause reasoning
Patient responsibility	Patient estimate reconciliation
Trace / EFT number	Payment reconciliation

19. Prior Authorization (278 / FHIR PAS)

Prior authorization is the **least-automated** workflow — only 40% electronic. Clearinghouse 278 support is **not universal**; some vendors (Availity, Optum) offer 278 APIs, others (Stedi) surface only the PA *requirement* via the 271. For CMS-regulated payers this moves to FHIR (Da Vinci PAS/CRD/DTR) by Jan 1, 2027.

Prior authorization (278 / FHIR PAS)



Da Vinci IG	Role
CRD — Coverage Requirements Discovery (v2.2.1)	At order time (CDS Hooks): does this need PA / documentation?
DTR — Documentation Templates and Rules (v2.2.0)	Smart payer questionnaires auto-populated from the EHR
PAS — Prior Authorization Support (v2.1.0)	Submits the PA request/response over FHIR; may map to X12 278 behind the scenes

Limitation & build

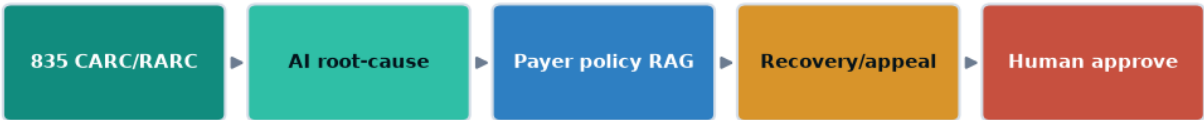
No single channel solves PA universally. We orchestrate: CRD (needed?) → DTR (docs) → 278/PAS submit → payer decision → portal/voice follow-up. We build the orchestration + documentation logic; we buy/partner for connectivity where it exists.

Sources: CMS-0057-F Interoperability & Prior Auth Final Rule (High) · HL7 Da Vinci PAS (Prior Auth Support) (High) · 2025 CAQH Index (adoption / savings) (Med-High)

20. Denials

Denial *data* comes from the 835 (CARC/RARC, adjustments) and 277CA (front-end rejections). Denial *intelligence* — root cause, recovery path, appeal likelihood — is ours to build. Payer medical policy, medical documentation and human review are still required for many denials.

Denial workflow (835 → reasoning → appeal)



Available electronically	Still requires
CARC/RARC codes (835)	Payer medical policy interpretation
Adjustment/payment amounts (835)	Medical documentation (records/notes)
Front-end rejects (277CA)	Human review for complex/clinical denials
Status (277)	Appeal drafting & submission (no universal API)

21. Appeals

There is no universal clearinghouse appeals API. Appeals are payer-specific: proprietary portals, fax, mail or payer web forms. This is a structural gap the platform must fill with its own workflow, AI drafting and cross-channel tracking.

Appeal channel	Reality
Electronic / API	NO universal standard; a few payer-specific portals
FHIR	No published appeals IG
Portal	Common — RPA/manual
Fax / mail	Still widespread
Documentation	Payer-specific packets
Status tracking	Manual; we build it

What we build

Appeal workflow, AI-drafted appeal letters (human-approved), documentation assembly, deadline tracking and cross-channel status — because no vendor provides it.

Sources: CMS Adopted Standards & Operating Rules (High) · X12 Health Care Transaction Flow (High)

22. Payer Policies

Clearinghouses do **not** provide medical/coverage policies, PA rules, documentation requirements, denial rules, timely-filing or appeal rules. Our **Payer Intelligence engine** collects and maintains these from payer manuals, websites, provider portals, FHIR/CRD, documents and human research — indexed for RAG so the platform can cite the rule behind every recommendation.

Policy type	Source we collect from
Medical / coverage policies	Payer manuals & websites; NCD/LCD for Medicare
PA rules	Payer PA lists; Da Vinci CRD (2027)
Documentation requirements	Payer packets; DTR questionnaires
Timely-filing & appeal rules	Payer manuals; contracts

23. Payer API

A **direct payer API** is a programmatic interface to one payer; a **clearinghouse API** is one interface to many. Direct payer APIs are better when a payer exposes richer, real-time or PA-specific data than the clearinghouse relays — but they are rare, uneven and add per-payer integration cost.

	Clearinghouse API	Direct payer API
Reach	Thousands of payers, one integration	One payer per integration
Data richness	Standardized (lowest common denominator)	Sometimes richer/real-time
Effort	Low (one vendor)	High (per payer)
When to use	Default for standard transactions	Selective, high-value payers / PA / FHIR

24. FHIR

FHIR (HL7 R4, REST + OAuth2 + SMART) is the future of payer connectivity for prior auth and patient/provider data. CMS-0057-F requires CMS-regulated payers to expose Prior Authorization, Patient Access, Provider Access and Payer-to-Payer FHIR APIs, generally by **Jan 1, 2027**. Note: the Patient Access API is **patient-authorized** — not a general provider RCM feed.

CMS rule	Summary
CMS-0057-F — Interoperability & Prior Authorization Final Rule (Jan 2024)	MA, Medicaid/CHIP FFS & managed care, QHP issuers on FFEs must build 4 FHIR R4 APIs: Prior Authorization (Da Vinci PAS/CRD/DTR), Patient Access (+PA info), Provider Access, Payer-to-Payer. API build generally by Jan 1, 2027. PA decision windows: 72h expedited / 7 days standard; specific denial reason required from 2026. Enforcement discretion: an all-FHIR PA API need not also use X12 278.
CMS-9115-F — Interoperability & Patient Access Final Rule (May 2020)	Patient Access API on FHIR R4 — PATIENT-AUTHORIZED data (CARIN Blue Button EOB); NOT a provider RCM feed. Provider Access API (provider-directed) is the separate 2027 requirement.
CMS-0053-F — Claims Attachments & Electronic Signatures (issued Mar 20, 2026; eff. May 26, 2026)	Adopts X12 v006020 275 (006020X314) & 277 (006020X313) as HIPAA attachment standards + HL7 C-CDA/Attachments IGs and an e-signature standard. Standardizes the historically fax/portal attachment workflow.
CMS-4201-F — 2024 MA & Part D Final Rule (Apr 2023) + Feb 2024 FAQ	MA medical-necessity/coverage decisions must reflect the individual's circumstances (NCD/LCD). An algorithm deciding coverage from a broad dataset (not the individual) is non-compliant — AI may assist but cannot replace individualized human medical-necessity determinations.
CA SB 1120 — 'Physicians Make Decisions Act' (eff. Jan 1, 2025)	Denial/delay/modification of care based on medical necessity must be made by a licensed physician/qualified provider with relevant expertise; AI may assist but cannot be the sole authority.

Sources: CMS-0057-F Interoperability & Prior Auth Final Rule (High) · CMS-9115-F Interoperability & Patient Access (High) · CMS-0053-F Claims Attachments Final Rule (2026) (High) · HL7 CARIN Blue Button (Patient Access) (High)

25. Portal

Payer portals still matter because for many workflows no EDI/API/FHIR path exists — prior-auth follow-up, attachment upload, appeal submission and status, and payer-specific tasks. Portal automation (RPA) is the next channel down, but it is brittle and must be built with guardrails, only where a digital path is genuinely absent.

26. AI Voice

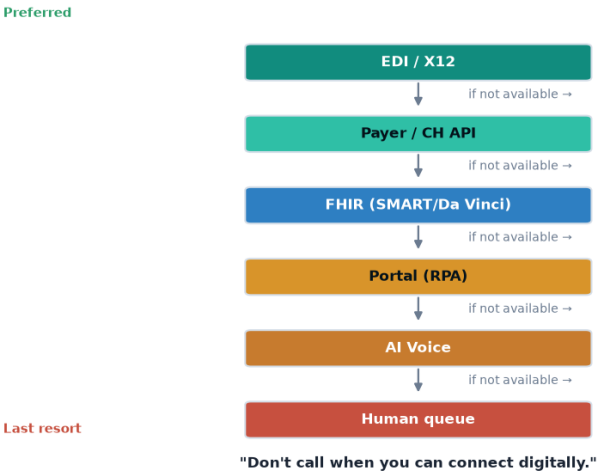
AI voice is useful for claim-status follow-up, authorization follow-up, missing-information calls, appeal follow-up and provider-services conversations — but it should **never be the first channel**. It is slow and costly relative to a digital transaction; it earns its place only when EDI, API, FHIR and portal cannot resolve the task.

AI voice fallback (only when digital fails)



27. Connectivity Orchestration

Our differentiator is the orchestration model: **“Don’t call when you can connect digitally.”** For every workflow the connectivity engine walks the ladder EDI → API → FHIR → Portal → AI Voice → Human, choosing the cheapest channel that can resolve the task.



Step	Channel	Use when
1. EDI / X12	Standardized, cheapest, highest-volume: eligibility, claims, status, ERA	Preferred for all HIPAA-mandated transactions
2. Payer / clearinghouse API	REST/JSON convenience layer over EDI, or direct payer API where richer	When EDI insufficient or a richer direct API exists
3. FHIR	SMART-on-FHIR / Da Vinci for prior auth (PAS/CRD/DTR) and payer data (2027)	For CMS-regulated payers and modern payer APIs
4. Payer portal (RPA)	Web automation when no EDI/API/FHIR path exists	Fallback for auth follow-up, attachments, appeals status
5. AI Voice	Automated phone call to payer IVR/rep for status/auth/missing-info	Only when no digital channel resolves it
6. Human work queue	Exception handling, judgment calls, complex appeals	Last resort; most expensive per action

28. Architecture

The AI Revenue Recovery connectivity engine routes across clearinghouse, payer APIs and FHIR; falls back to portals, AI voice and humans; captures the outcome; and drives the AI workflow to a revenue action — all under an AI connectivity orchestrator with a shared intelligence, audit and tenant-isolation plane.

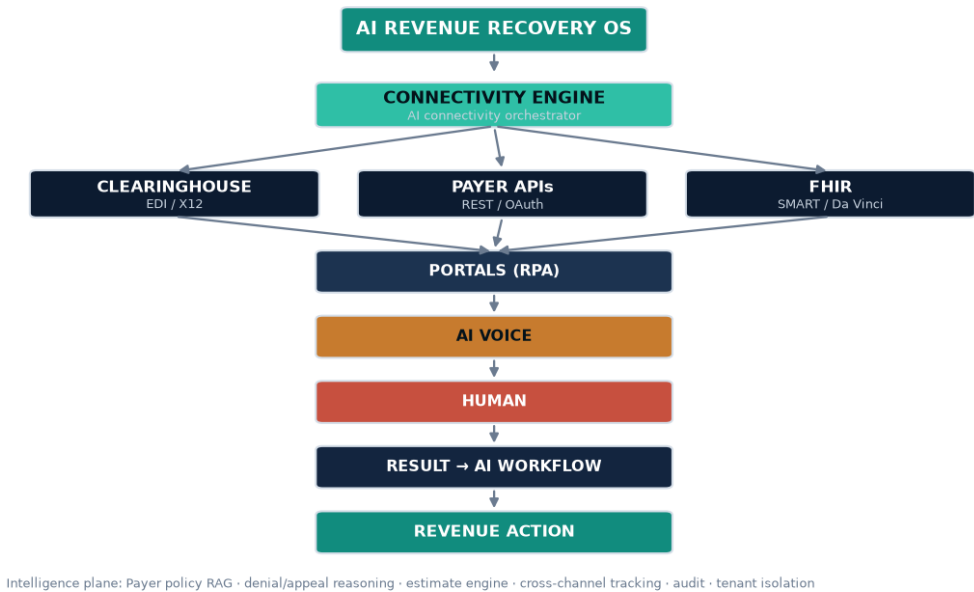
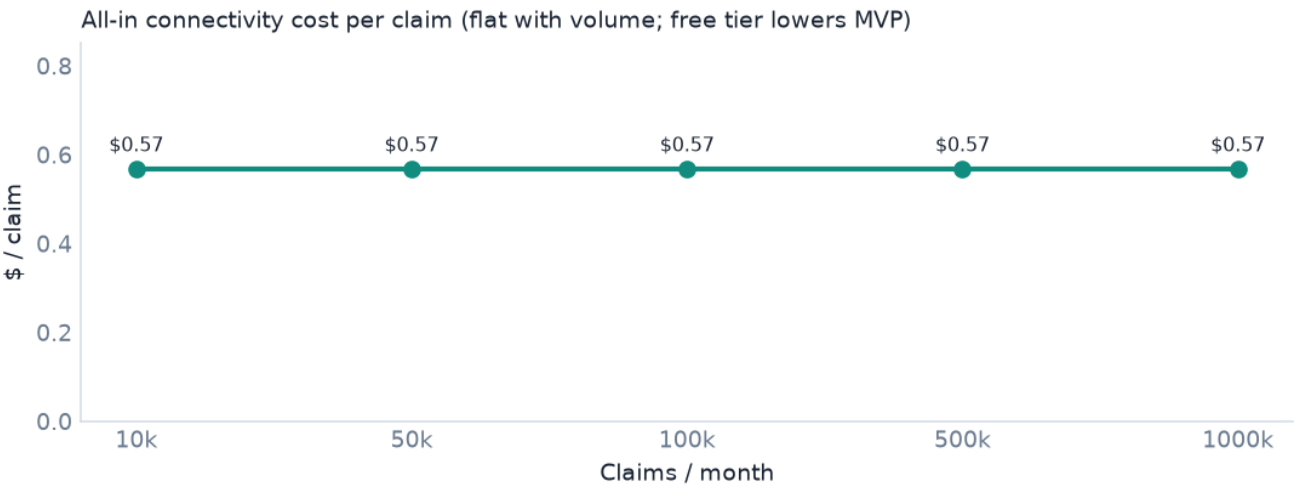
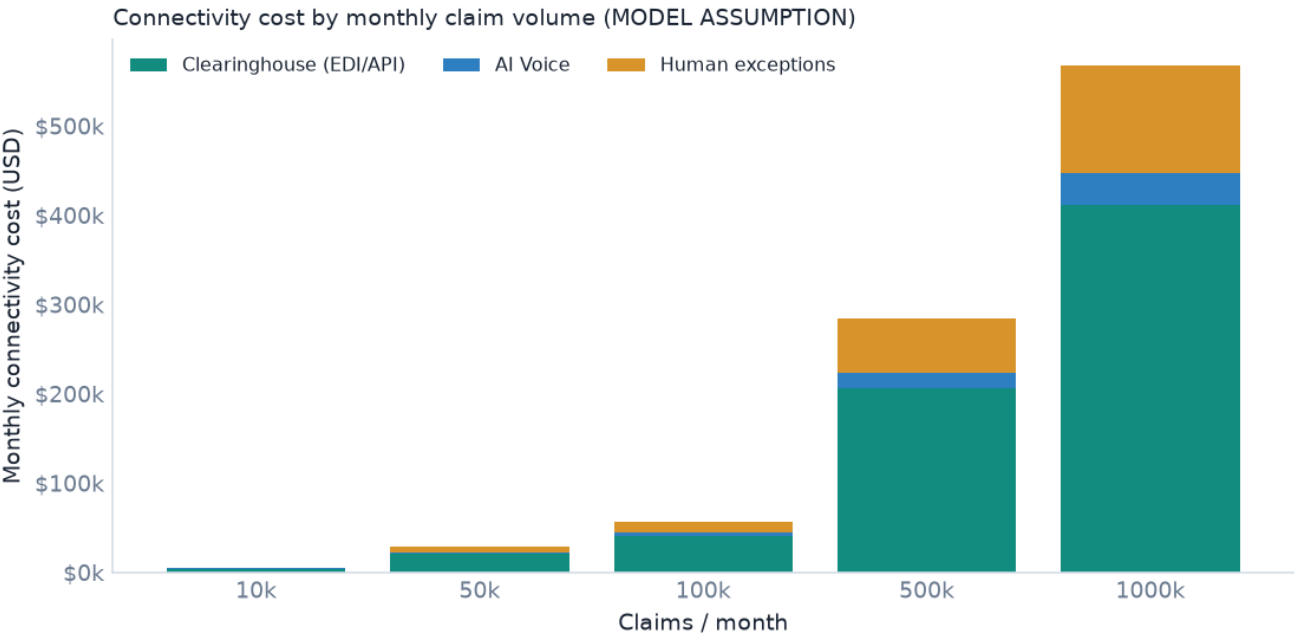


Figure. AI Revenue Recovery OS connectivity engine and orchestration ladder.

29. Cost Model

A parameterized connectivity cost model across five volume scenarios. Clearinghouse per-transaction rates are usage-tiered / QUOTE REQUIRED, so unit costs here are **MODEL ASSUMPTIONS**; voice and human costs dominate the tail because they are the most expensive channels per action.



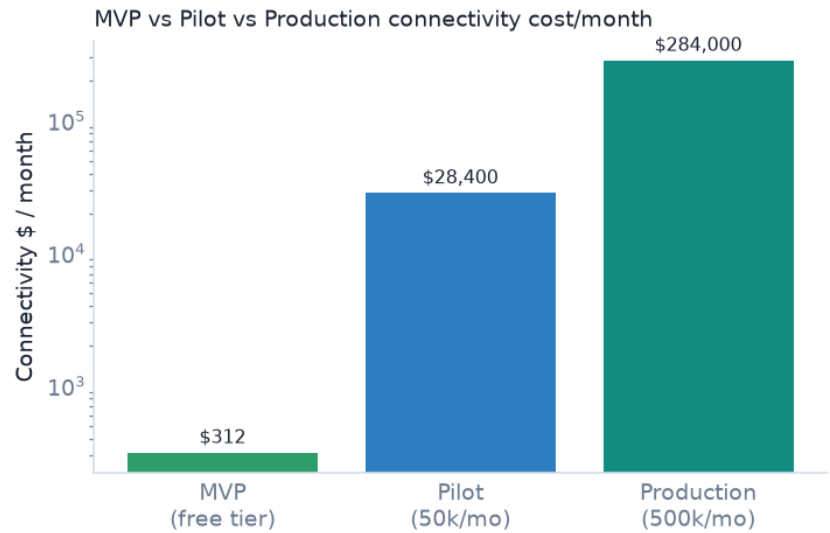
Claims/mo	Clearinghouse	Voice	Human	Total/mo	\$/claim	Annual
10,000	\$4,120	\$360	\$1,200	\$5,680	\$0.568	\$68,160
50,000	\$20,600	\$1,800	\$6,000	\$28,400	\$0.568	\$340,800
100,000	\$41,200	\$3,600	\$12,000	\$56,800	\$0.568	\$681,600
500,000	\$206,000	\$18,000	\$60,000	\$284,000	\$0.568	\$3,408,000
1,000,000	\$412,000	\$36,000	\$120,000	\$568,000	\$0.568	\$6,816,000

Channel economics

Cost rises sharply down the ladder — a digital transaction is cents; a voice call is dollars; a human touch is several dollars. Every workflow the orchestrator keeps digital protects margin.

30. MVP vs Pilot vs Production

A free MVP is feasible: build the investor demonstration on free sandboxes, mock payer APIs, synthetic data, test EDI and developer accounts — paying no production transaction fees. Pilot and production then move to paid usage.



Phase	Volume	Connectivity \$/mo	Basis
MVP / demo	~2,000 (free tier)	~\$312	Free sandbox + Stedi Basic (100 free txns); synthetic data
Pilot	50,000	~\$28,400	First paid production; QUOTE-based unit rates
Production	500,000	~\$284,000	Medium production; verify live tiered pricing

Free MVP strategy

Stedi free sandbox + Basic plan, Availity Demo, synthetic/de-identified data and mock payer responses let us build and demo the full orchestration flow at ~\$0 connectivity cost — no PHI, no production fees.

31. Build vs Buy vs Partner

Buy commodity connectivity; build the intelligence and orchestration; partner for voice. The rule of thumb: if it is regulated, standardized and undifferentiated, buy it; if it is where our advantage lives, build it.

Capability	Recommendation	Rationale
EDI / X12 engine & payer routing	BUY (clearinghouse)	Commodity, regulated, high-maintenance; clearinghouses do this at scale
Eligibility / benefits (270/271)	BUY (clearinghouse)	Universal clearinghouse capability; no edge in rebuilding
Claim submission (837) & ERA (835)	BUY (clearinghouse)	Core clearinghouse rails + payer enrollment handled
Claim status (276/277)	BUY (clearinghouse)	Widely available via CH APIs
Insurance / coverage discovery	BUY / PARTNER	Available as paid add-on (Waystar/Experian/Optum); buy vs build by ROI
Prior auth connectivity (278/FHIR PAS)	PARTNER + BUILD	Use CH/payer where available; build orchestration + DTR logic
Patient responsibility estimate	BUILD	Our calculation engine over CH benefit data — differentiator
Denial intelligence (CARC/RARC reasoning)	BUILD	AI reasoning over 835/277CA is core IP
Appeal intelligence & drafting	BUILD	No universal API; AI drafting + tracking is core IP
Payer policy intelligence (RAG)	BUILD	Collect/maintain payer rules; not a clearinghouse function
FHIR gateway (SMART/Da Vinci)	BUILD / PARTNER	Build the gateway; partner for niche IGs
Portal automation (RPA)	BUILD carefully	Brittle; build with guardrails only where no digital path exists
AI Voice	PARTNER	Use a voice platform (Retell/Vapi/Twilio-based); orchestrate, don't rebuild telephony
Workflow orchestration & connectivity engine	BUILD	The differentiated orchestration layer — our product
Analytics / audit / tenant isolation	BUILD	Product-specific; security-critical

32. Product Opportunity

The opportunity is **not** to become another clearinghouse. It is to build the AI intelligence and orchestration layer **above** existing healthcare connectivity — working with multiple clearinghouses, direct payer APIs, FHIR, portals, voice and human workflows.

Clearinghouses solve (we buy)	We build (our moat)
EDI connectivity and X12 envelope handling	Payer intelligence & coverage-policy RAG
Payer routing across thousands of payer connections	AI reasoning and denial analysis (CARC/RARC → root cause → recovery)
Eligibility & benefits (270/271)	Appeal intelligence, drafting and tracking (no universal API)
Claim submission (837P/I/D) and payer enrollment plumbing	Patient responsibility estimate engine
Claim status (276/277)	Connectivity orchestration (EDI→API→FHIR→Portal→Voice→Human)
Electronic remittance (835 / ERA)	Portal automation orchestration (guarded RPA)
Acknowledgments (277CA / 999 / TA1)	AI voice fallback orchestration
	Cross-channel tracking and outcome capture
	Human exception management
	Analytics, audit, and multi-tenant isolation
	Its own HIPAA-aligned security & compliance program

33. Security / HIPAA

A clearinghouse's security does **not** make our application HIPAA compliant. We run our own HIPAA-aligned program: signed BAAs with every clearinghouse and cloud provider handling PHI, encryption in transit and at rest, least-privilege access, tenant isolation, API security (OAuth2, secrets management) and immutable audit logging.

Control	Our responsibility
BAA	Signed with each clearinghouse/cloud handling PHI
Encryption	TLS 1.2+ in transit; AES-256 at rest; managed keys/HSM
Access control	Least privilege; MFA; RBAC; tenant scoping
Tenant isolation	RLS → schema → dedicated; provider vs payer never mixed
API security	OAuth2, key rotation, secrets vault, rate limiting
Audit & logging	Immutable logs of every access & AI decision

HIPAA posture

HIPAA-**aligned** architecture and compliance program — never “HIPAA certified.” Vendor certifications (e.g. Stedi SOC 2 Type II / HIPAA-eligible) are inputs, not a substitute for our own program.

Sources: Stedi SOC 2 Type II & HIPAA eligibility (High) · CMS-4201-F 2024 MA & Part D Final Rule (High) · CA SB 1120 Physicians Make Decisions Act (High)

34. Investor Summary

#	Question	Answer
1	Is clearinghouse API free?	Partly — free sandbox & small free tiers; production usually usage-based/quote
2	What is free?	Stedi Basic 100 txns/mo; Availity Essentials (par payers); dev sandboxes
3	What is paid?	Production at scale — usage tiers or QUOTE REQUIRED
4	What can clearinghouses cover?	Eligibility, claims, status, ERA, acks; attachments emerging
5	What can't they cover?	Universal PA, appeals, payer policies, denial reasoning, orchestration
6	How many payers?	Thousands via one integration (3,500–8,000+ connections)
7	Need payer APIs?	Yes — selectively, for richer/PA data
8	Need FHIR?	Yes — PA (Da Vinci) & CMS payers by 2027
9	Need portals?	Yes — fallback where no EDI/API/FHIR
10	Need AI voice?	Yes — last digital-fallback before humans; never first
11	What to build?	Intelligence + orchestration + estimate/denial/appeal engines
12	What to buy?	EDI/clearinghouse connectivity
13	What to partner?	AI voice; niche FHIR
14	MVP connectivity cost	~\$312/mo (free/sandbox)
15	Pilot connectivity cost	~\$28,400/mo (50k claims)
16	Production connectivity cost	~\$284,000/mo (500k claims)

35. Final Recommendation

Do not build a clearinghouse. Build an AI Revenue Recovery orchestration platform that uses clearinghouses as one connectivity layer — alongside direct payer APIs, FHIR, portals, AI voice and human workflows — controlled by an AI connectivity orchestrator.

- **Start provider-side** on clearinghouse APIs; a near-\$0 MVP is feasible on free/sandbox tiers with synthetic data.
- **Stay multi-vendor** — abstract clearinghouses behind our connectivity engine so no single vendor (or its pricing) locks us in.
- **Invest the moat** in payer intelligence, denial/appeal reasoning, patient-estimate logic and cross-channel orchestration — the parts no clearinghouse sells.
- **Add payer-side** capabilities as decision-support after validation, with a qualified clinician owning every adverse medical-necessity decision.
- **Verify live pricing** before board commitment — most production rates are usage-tiered or QUOTE REQUIRED.

Strategic conclusion

Clearinghouse + Direct Payer APIs + FHIR + Portal + AI Voice + Human — controlled by an AI Connectivity Orchestrator. The clearinghouse solves connectivity; we own the intelligence and orchestration above it. That is a differentiated platform, not another clearinghouse and not just an AI-calling app.

Appendix. Sources

Every pricing, coverage and API-availability claim is tied to a source. Full URLs are hyperlinked in the workbook SOURCES sheet.

Source	Date	Confidence
Stedi pricing / docs / billing	2026-08-20	High
Stedi Basic plan (100 free txns/mo)	2026-08-20	High
Stedi pay-as-you-go pricing	2026-08-20	High
Stedi test mode / sandbox	2026-08-20	High
Stedi transaction enrollment	2026-08-20	High
Stedi SOC 2 Type II & HIPAA eligibility	2026-08-20	High
Availity Developer Portal getting started	2026-08-20	High
Availity Essentials (free portal)	2026-08-20	High
Availity HIPAA transactions (APIs)	2026-08-20	High
Availity network size (Apr 2026)	2026-04-28	High
Waystar platform (financial clearance/denials)	2026-08-20	High
Waystar Coverage Detection	2026-08-20	High
Waystar Hubble AI platform	2026-08-20	High
Optum developer — access the APIs	2026-08-20	High
Optum Enhanced Eligibility (Coverage Discovery)	2026-08-20	High
Optum Prior Authorization Submission API	2026-08-20	High
Optum Medical Network EDI (payer count)	2026-08-20	Med
Change Healthcare cyberattack recovery	2026-08-20	High
Experian Health eligibility (900+ payers)	2026-08-20	High
Experian Health AI Advantage denials	2023-11-30	High
Office Ally pricing (Par/Non-Par)	2026-08-20	High
Office Ally Par vs Non-Par fees (\$44.95)	2026-08-20	High
Claim.MD quick-start / pricing	2026-08-20	Med
TriZetto Provider Solutions clearinghouse	2026-08-20	Med
X12 Health Care Transaction Flow	2026-08-20	High
CMS Adopted Standards & Operating Rules	2026-08-20	High
CMS-0057-F Interoperability & Prior Auth Final Rule	2026-08-20	High
CMS-9115-F Interoperability & Patient Access	2026-08-20	High
CMS-0053-F Claims Attachments Final Rule (2026)	2026-03-24	High
CMS-4201-F 2024 MA & Part D Final Rule	2026-08-20	High
CA SB 1120 Physicians Make Decisions Act	2026-08-20	High
HL7 Da Vinci PAS (Prior Auth Support)	2026-08-20	High
HL7 Da Vinci CRD	2026-08-20	High
HL7 Da Vinci DTR	2026-08-20	High
HL7 CARIN Blue Button (Patient Access)	2026-08-20	High
2025 CAQH Index (adoption / savings)	2026-02-19	Med-High
AI voice per-minute (blended estimate)	2026-08-20	Est